# School of Information Technology, Engineering, Mathematics and Physics

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**CS241 – Software Design & Implementation**
**Semester 2, 2022**
**Mode: Face to Face**

**Short Test 1**
**Date: Friday 30 September 2022, Time: 9.00 – 9.50pm (FJT)**

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*Time Allowed: 50mins*
*40 marks (Weight: 10%)*

**Student ID: _____**
**Student Name: _____**

**LEARNING OUTCOMES:**

- Develop software design document
- Develop a simple software system based on the design document

**INSTRUCTIONS**

1) This exam has three sections:
   a) Section A: Short-answer questions (15 marks)
   b) Section B: Long answers/Software Design Artefacts (15 marks)
   c) Section C: Programming (10 marks)
2) Answer all questions in this booklet. All questions are compulsory.
3) The total number of pages including this cover sheet is 6. Extra answer sheet is also attached.

**~ALL THE BEST~**

## *Section A: Short Answer Questions* *(15 Marks)*

## Question 1 (5 marks)

a) The waterfall model is a classical model used in system development life cycle to create a system with a linear and sequential approach. Discuss at least one advantages and 1 disadvantage of such model. (2 marks)

Advantages

- Before the next phase of development, each phase must be completed
- Suited for smaller projects where requirements are well define
- Project is completely dependent on project team with minimum client intervention

Disadvantages

- Error can be fixed only during the phase
- It is not desirable for complex project where requirement changes frequently

a) Clients valuable feedback cannot be included with ongoing development phase

OR Any further relevant Advantages or Disadvantaged, please award marks.

b) Team organization is very important in Software Engineering. What is democratic team approach and give a strength and challenge with this approach. (3 marks)

Egoless programming, Restructure the social environment, Restructure programmers' values

Difficulty: Management may have difficulties, Democratic teams are hard to introduce into an undemocratic environment

Strengths:
Democratic teams are enormously productive
They work best when the problem is difficult
They function well in a research environment

## Question 2 (5 marks)

a) Software engineering ethics is a type of professional ethics. Software engineering is often a profession practiced by teams in managed situations, so as we shall see a complete concept of software engineering ethics includes ethical activity and principles related to the action of teams and the actions of management. Describe any two issues of professional responsibility of the software engineering as part of Software engineering ethics. (2 marks)

- Confidentiality

- Intellectual property rights
- Computer misuse
- Competence

b) In Object Oriented Analysis and Design, the three class types in the Unified Process are Entity classes, Boundary classes and Control classes. With example explain these three class types. (3 marks)

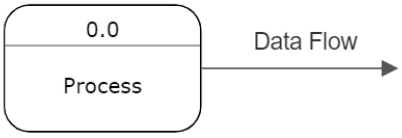
- An entity class models information that represents a business concept such as customer, employee, student contain useful information. Eg Account class
- A boundary class models the interaction between the information system and its actors. Boundary classes are generally associated with input and output. User interface classes are boundary class. Eg report class
- A control class models complex computations and algorithms. Eg calculation fees

## Question 3 (5 marks)

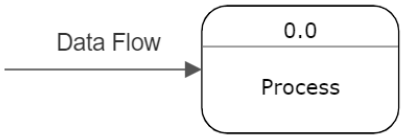
a) One of the phase in the SDLC phases is the Design phase. Explain what happens in the design phase and state it's importance. (2 marks)

The purpose of the Design Phase is to transform the requirements into complete and detailed system design specifications. Once the design is approved, the Development Team begins the Development Phase. OR Relevant explanations accepted.

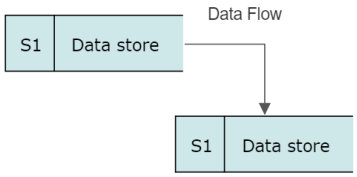
b) In classical analysis and design, Data-flow model is one of the system models that describes the overall behavior of the system especially data processing in the system. There are certain rules for governing construction of Data Flow Diagrams. For the following incorrect diagrams, specify the rule or explain what is incorrect in each. (3 marks)




A process cannot have only outputs (Miracle)



A process cannot have only inputs (Black hole)




All data stores must be connected to at least one process.

## *Section B: Long answers/Software Design Artefacts* *(15 Marks)*

## Question 4 (4 mark)

a) Explain the differences between the structured and object-oriented approaches to systems development. (2 marks)

Structured (traditional) approach – based on techniques used to develop information systems with structured and modular programming. Object-oriented approach – views an information system as a collection of interacting objects that work together to accomplish tasks.
Note: The difference between these two approaches of software development is defined by how data and processes of the system are handled. In a structured approach, both data and processes are handled separately leading to the system being developed as a set of modules. In an object-oriented approach, both data and processes are models in a single unit called classes.

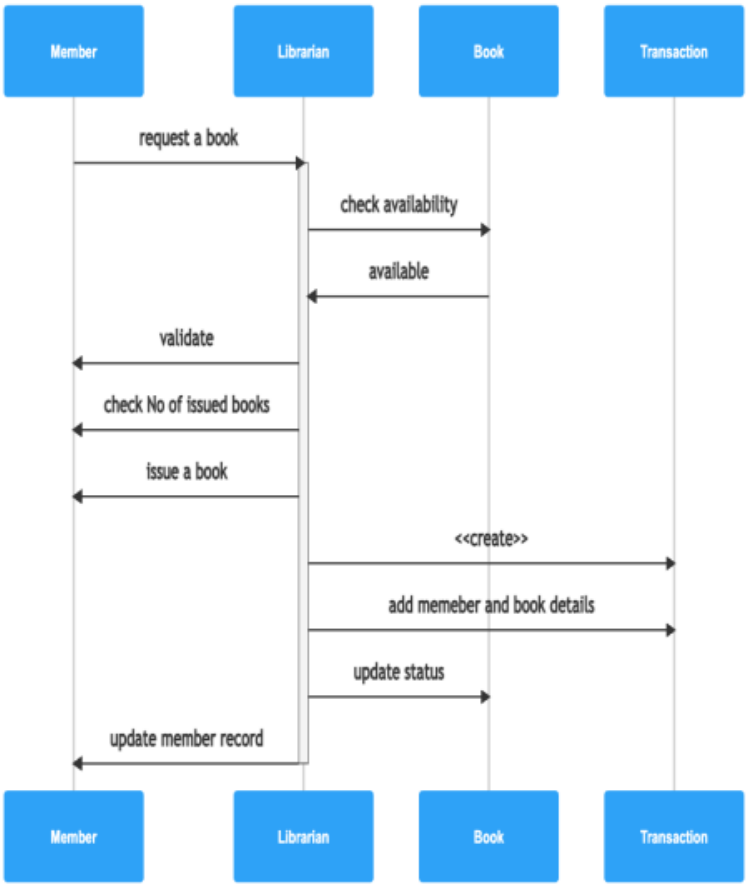
b) Explain any two main maintenance activities. (2 marks)
- Corrective maintenance
- Fixing faults
- Enhancement (Software update)
- Implement changes in specification
- Perfective maintenance
- Adding functionality or making system more efficient
- Adaptive maintenance
- Making changes because the environment changes such as new hardware/operating system.

## Question 5 (5 mark)

**Library management system (LMS) – (Scenario)**
The library management system is used by the library to manage the issuing and borrowing of the books by library member. First, the member requests a book from the librarian where the librarian checks the availability of the book and validates the request. Before issuing the book, the librarian checks the number of books the member has checked out. Then the member gets the requested book. The librarian creates the transaction and documents the member and book

details. After the librarian creates the transaction, the librarian updates the status of the issued book and member record.

*Create a complete sequence diagram for the above scenario.*



## Question 6 (6 mark)

Data flow diagrams (DFDs) are decomposed into additional diagrams to provide multiple levels of detail. Higher-level diagrams provide general views of system and lower-level diagrams provide detailed views of system starting with level 0.

The figure below shows a context diagram for Recruitment System which is used by the company for recruitment. The system is used by Applicants to apply for any job openings and get the interview schedule if they get shortlisted. The Software Engineering Manager uses the system to hire new staff by providing job descriptions to the system for new openings and they also get the interview schedule for the interviews with shorted listed applicants.

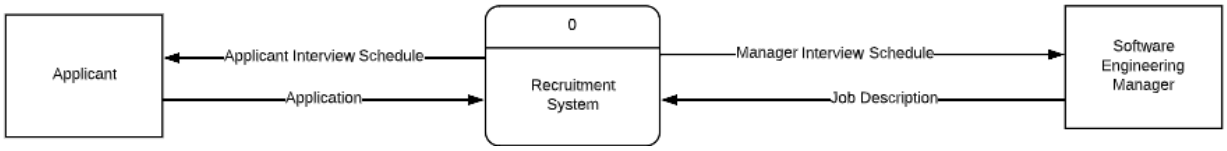



*Figure 1: Context diagram for Recruitment System*

*Create the Level 0 DFD diagram for the Recruitment System.*

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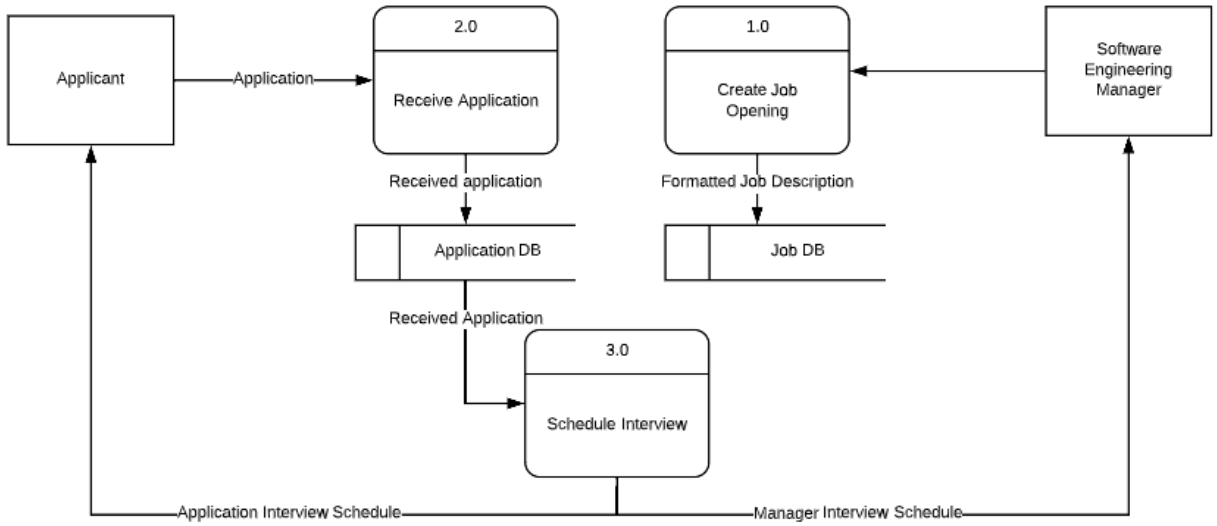
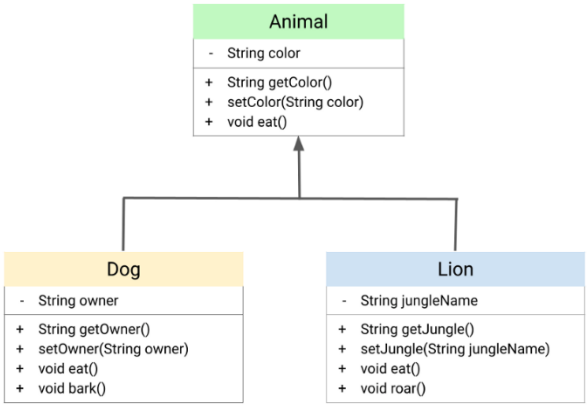



## *Section C: Programming* *(10 Marks)*

## Question 6 (5 mark)

Inheritance is another representation of an OO relationship between two classes. Use the class diagram below to answer the following questions.




a) State the base and derived classes. (2 marks)

Base: Animal

Derived: Dog and Lion

b) List the Generic, Specific and Unique methods from the above classes. (3 marks)

Generic: getColor(), setColor(),

Specific: eat()

Unique: getOwner(), setOwner(), bark(), getJungle(), setJungle(), roar()

## Question 7 (5 mark)

***Use the following code snippet to answer the questions that follows on the right below:***

```java
class Car {
  int year = 0;
  public Car(){
    System.out.println("Car constructor");
  }
  public Car(int year){
    this.year = year;
    System.out.println("Car year: " + this.year);
  }
  public void carSound() {
    System.out.println("Car sounds: boom boom");
  }
}
class BMW extends Car {
  public BMW() {
    System.out.println("BMW constructor");
  }
  public void carSound() {
    System.out.println("BMW sounds: zee zee");
  }
}

class Main {
  public static void main(String[] args) {
    Car myCar = new Car(3);
    BMW myBMW = new BMW();

    myCar.carSound();
    myBMW.carSound();
  }
}
```

Assume that all the imports and packages are correct and code is executing without any error.

a) Once the program executes, write what will be printed. Write in exact order. (4 marks)

Car year: 3
Car constructor
BMW constructor
Car sounds: boom boom
BMW sounds: zee zee

b) Give one example of overridden method in this code. (1 mark)

**THE END**

**EXTRA ANSWER SHEET – USE ONLY WHEN NEEDED**

- Mark the question number properly if used.